

NTRK1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK1 gene fusions from the msk_impact_50k_2026 study, covering 78 fusion events. 32/78 fusions (41.0%) were in-frame. 64/78 fusions (82.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0482628$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 78 analyzed fusion events, identifying 43 distinct fusion partner genes. The most frequent partner was LMNA, observed in 12 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 30/32 (93.8%) of in-frame fusions retained the domain, $p=0.0482628$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 31/32 (96.9%) of in-frame fusions disrupted the domain, $p=0.0744325$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.492351.

Cutpoint detection

Cutpoint detection scanned 76 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 501 aa (permutation-corrected $p=0.014985$, statistically significant at $\alpha=0.05$). This is -11 aa from the nearest configured domain boundary at 512 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=64$) and "not_retained" ($n=12$). For Frame_status, retained: 30/36 (83.3%, 95% CI 68.1%-92.1%); not_retained: 2/8 (25.0%, 95% CI 7.1%-59.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 66.3594 vs 43.25, $p=0.200314$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (43 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: NTRK1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
LMNA	12	0.15384615384615385	0.30004340774793187	0.030772538301758395	0.852906976744186	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.31728281105779194	1
TPM3	12	0.15384615384615385	0.30004340774793187	0.030772538301758395	0.7467054263565891	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3013525784996524	2
MEF2D	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	1.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2970390722384717	3
TRPM8	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	1.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2970390722384717	4
ARHGEF11	2	0.02564102564102564	0.30004340774793187	0.030772538301758395	0.9430232558139535	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29233871445671855	5
RABGAP1L	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27994604898265774	6
VANGL2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27994604898265774	7
ZBTB7B	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27994604898265774	8
SCP2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8651162790697674	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2768065140989368	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TPR	5	0.0641025641025641	0.30004340774793187	0.030772538301758395	0.7427906976744186	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2738422922742499	10
LTAP1	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8372093023255813	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2726204675873089	11
ATP1A2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8186046511627907	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2698297699128903	12
PLEKHA6	3	0.038461538461538464	0.30004340774793187	0.030772538301758395	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26914998458194217	13
IRF2BP2	3	0.038461538461538464	0.30004340774793187	0.030772538301758395	0.758139534883721	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26845231016333754	14
ANKRD36	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8069767441860465	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2680855838663787	15
RAB25	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.8046511627906977	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26773674665707636	16
IQGAP3	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7976744186046512	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2666902350291694	17
SMYD2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7767441860465116	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26355070014544846	18
EML4	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2614576768896345	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EPS15	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2614576768896345	20
F11R	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2614576768896345	21
PEAR1	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2614576768896345	22
AFAP1	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25936465363382055	23
DIAPH1	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25936465363382055	24
TRIM63	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25936465363382055	25
STK11	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7325581395348837	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25692279316870426	26
KIF21B	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7255813953488373	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25587628154079733	27
GON4L	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.7162790697674419	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.254480932703588	28
BCAN	2	0.02564102564102564	0.30004340774793187	0.030772538301758395	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22518755166602092	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CTRC	2	0.02564102564102564	0.30004340774793187	0.030772538301758395	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22518755166602092	30
NOS1AP	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.5186046511627906	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2248297699128903	31
NELFCD	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22134139781986706	32
SEMA4A	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22134139781986706	33
COP1	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.4534883720930233	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2150623280524252	34
TARS2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.2697674418604651	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18750418851754147	35
P2RY8	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.2581395348837209	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18576000247102986	36
DDR2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	None	0.6202755879498323	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.17298714380996674	37
METTL25B	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	None	0.6202755879498323	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.17298714380996674	38
SLAMF6	3	0.038461538461538464	0.30004340774793187	0.030772538301758395	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17182440318659337	39
CADM3	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16413209549428567	40

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRCC	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16413209549428567	41
TAF2	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16413209549428567	42
SHPRH	1	0.01282051282051282	0.30004340774793187	0.030772538301758395	0.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.14703907223847174	43

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
71	0	64	1	11	0.0	0.15789473684210525	0.8016323462331666
120	5	59	2	10	0.423728813559322	0.3043106841520683	0.5166827995746912
182	5	59	3	9	0.2542372881355932	0.10748982443269033	0.9686326464637113
187	6	58	3	9	0.3103448275862069	0.14608214986786255	0.83540284831964
284	10	54	5	7	0.25925925925925924	0.05253839246726117	1.2795232195834179
294	11	53	5	7	0.29056603773584905	0.11464025310482523	0.9406628637664377
321	13	51	5	7	0.3568627450980392	0.14136093824402443	0.8496705809825442
343	14	50	5	7	0.392	0.1614921032975089	0.7918487091109843
359	15	49	5	7	0.42857142857142855	0.28166616270038797	0.550265322823117
377	16	48	5	7	0.4666666666666667	0.29450254706111656	0.530910944782467
379	17	47	5	7	0.5063829787234042	0.31247043251825074	0.505191071445305
383	18	46	5	7	0.5478260869565217	0.494013437398657	0.30626123790080845
393	27	37	5	7	1.0216216216216216	1.0	-0.0
399	39	25	5	7	2.184	0.33960073354007925	0.46903138034780045
405	39	25	6	6	1.56	0.5328211446391313	0.27341854881920497
411	40	24	6	6	1.6666666666666667	0.5236156499522178	0.28098738162793063
414	41	23	6	6	1.7826086956521738	0.5184932375861125	0.2852569034951671
417	41	23	7	5	1.2732919254658386	0.7509447780617822	0.12439199836843534
418	48	16	7	5	2.142857142857143	0.29450254706111656	0.530910944782467

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
423	49	15	7	5	2.3333333333333335	0.28166616270038797	0.550265322823117
443	50	14	7	5	2.5510204081632653	0.1614921032975089	0.7918487091109843
452	61	3	8	4	10.166666666666666	0.010191376831302902	1.9917671398557228
492	62	2	8	4	15.5	0.004800736843751964	2.3186920995794114
501	64	0	9	3	None	0.0031294452347083927	2.5045326441976177
571	64	0	10	2	None	0.023157894736842103	1.6353009244666417
616	64	0	11	1	None	0.15789473684210525	0.8016323462331666

domain_disruption tables

frame_domain_contingency_table

31	37
1	7

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf13855	PF13855	76	8	6	62	0.045862412761714856
auto_disruption_pf13895	PF13895	76	13	0	63	0.0

domain_retention tables

frame_domain_contingency_table

30	34
2	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	76	64	3	9	0.1117283950617284

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
13	65	78	0.8333333333333334

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AFAP1	1
ANKRD36	1
ARHGEF11	2
ATP1A2	1
BCAN	2
CADM3	1
COP1	1
CTRC	2
DDR2	1
DIAPH1	1
EML4	1
EPS15	1
F11R	1
GON4L	1
IQGAP3	1
IRF2BP2	3
KIF21B	1
LMNA	12
LTAP1	1
MEF2D	1
METTL25B	1
NELFCD	1
NOS1AP	1
P2RY8	1
PEAR1	1
PLEKHA6	3
PRCC	1
RAB25	1
RABGAP1L	1

Partner_gene	Event_count
SCP2	1
SEMA4A	1
SHPRH	1
SLAMF6	3
SMYD2	1
STK11	1
TAF12	1
TARS2	1
TPM3	12
TPR	5
TRIM63	1
TRPM8	1
VANGL2	1
ZBTB7B	1

Per-event results (results.tsv)

event_id	sampl_e_id	pa_tie_int_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0013559-T01-IM5-1	P-0013559-T01-IM5		AFAP1-NTRK1	AFA1P1	in-frame	three_prime	156844093	9	393	True	retain_ed	1.0	False	Protein kinase domain	PF13855; P F13895		AFAP1(NM_001134647) - NTRK1(NM_002529) Rearrangement : t(1;4)(q23.1;p16.1)(chr1:g.156844093::chr4:g.7852375) Protein fusion: in frame (AFAP1-NTRK1)	NA
EVT-P-0042270-T01-IM6-2	P-0042270-T01-IM6		ANKRD36-NTRK1	ANKRD36	out-of-frame	three_prime	156844593	11	418	True	retain_ed	1.0	False	Protein kinase domain	PF13855; P F13895		ANKRD36 (NM_001164315) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;q11.2)(chr1:g.156844593::chr2:g.97835462) Protein Fusion: out of frame {ANKRD36:NTRK1}	NA
EVT-P-0017558-T02-IM7-3	P-0017558-T02-IM7		ARHGEF11-NTRK1	ARHGEF11	out-of-frame	three_prime	156845502	13	501	True	retain_ed	1.0	False	Protein kinase domain	PF13855; P F13895		ARHGEF11 (NM_198236) - NTRK1 (NM_002529) fusion: c.32+488:ARHGEF11_c.1501+44:NTRK1inv Protein Fusion: out of frame {ARHGEF11:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 0637-T01-I M6-4	P-0050 637-T0 1-IM6		BCAN- NTRK1	BCAN	in-frame	three_prime	156842373	8	284	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c.1064-525:BCAN_c.850+826:NTRK1del Protein Fusion: in frame {BCAN:NTRK1}	NA
EVT-P-004 1874-T01-I M6-5	P-0041 874-T0 1-IM6		BCAN- NTRK1	BCAN	in-frame	three_prime	156843258	8	284	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c.92-224:BCAN_c.851-167:NTRK1del Protein Fusion: in frame {BCAN:NTRK1}	NA
EVT-P-005 5952-T01-I M6-6	P-0055 952-T0 1-IM6		COP1- NTRK1	COP1	in-frame	three_prime	156851106	17	736	True	disrupted	0.170370370370370	True		PF13855; P F13895	Protein kinase domain	RFWD2 (NM_022457) - NTRK1 (NM_002529) rearrangement: c.2133+18462:RFWD2_c.2206-143:NTRK1inv Protein Fusion: in frame {RFWD2:NTRK1}	NA
EVT-P-001 4859-T03-I M6-8	P-0014 859-T0 3-IM6		CTRC- NTRK1	CTRC	unknown	three_prime	156841593	8	284	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+46:NTRK1del Protein Fusion: mid-exon {CTRC:NTRK1}	NA
EVT-P-001 4859-T01-I M6-9	P-0014 859-T0 1-IM6		CTRC- NTRK1	CTRC	unknown	three_prime	156841592	8	284	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+45:NTRK1del Protein fusion: mid-exon (CTRC-NTRK1)	NA
EVT-P-002 9326-T01-I M6-10	P-0029 326-T0 1-IM6		DIAPH1- NTRK1	DIAPH1	in-frame	three_prime	156843860	9	393	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		DIAPH1 (NM_005219) - NTRK1 (NM_002529) rearrangement: t(1;5)(q23.1;q31.3)(chr1:g.156843860::chr5:g.140904837) Protein Fusion: in frame {DIAPH1:NTRK1}	NA
EVT-P-005 9139-T01-I M7-11	P-0059 139-T0 1-IM7		EML4- NTRK1	EML4	in-frame	three_prime	156844256	10	399	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		EML4 (NM_019063) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;p21)(chr1:g.156844256::chr2:g.42473987) Protein Fusion: in frame {EML4:NTRK1}	NA
EVT-P-003 7901-T01-I M6-12	P-0037 901-T0 1-IM6		EPS15- NTRK1	EPS15	in-frame	three_prime	156844212	10	399	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		EPS15 (NM_001981) - NTRK1 (NM_002529) rearrangement: c.2119+1240:EPS15_c.1195+20:NTRK1inv Protein Fusion: in frame {EPS15:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009855-T01-I M5-13	P-0009855-T01-IM5		GON4L-NTRK1	GON4L	unknown	threep_rime	156843711	8	379	False	retained	1.0	False	Protein kinase domain	PF13855; P F13895		GON4L (NM_001037533) - NTRK1 (NM_002529) rearrangement : c.4141:GON4L_c.1137:NTRK1inv Protein fusion: mid-exon (GON4L-NTRK1)	NA
EVT-P-0050669-T01-I M6-15	P-0050669-T01-IM6		IQGAP3-NTRK1	IQGAP3	unknown	threep_rime	156844408	10	414	False	retained	1.0	False	Protein kinase domain	PF13855; P F13895		IQGAP3 (NM_178229) - NTRK1 (NM_002529) fusion: c.2531-131:IQGAP3_c.1241:NTRK1inv Protein Fusion: mid-exon {IQGAP3:NTRK1}	NA
EVT-P-0022925-T01-I M6-16	P-0022925-T01-IM6		LMNA-NTRK1	LMNA	unknown	threep_rime	156844398	10	411	False	retained	1.0	False	Protein kinase domain	PF13855; P F13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-3 fused to NTRK1 exons 10 - 17): c.513+1675:LMNA_c.1231:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0010581-T01-I M5-17	P-0010581-T01-IM5		LMNA-NTRK1	LMNA	in-frame	threep_rime	156844556	11	418	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 11-17): c.514-811:LMNA_c.1251+138:NTRK1del Protein fusion: in frame (LMNA-NTRK1)	NA
EVT-P-0010086-T01-I M5-18	P-0010086-T01-IM5		LMNA-NTRK1	LMNA	in-frame	threep_rime	156844436	11	418	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		LMNA (NM_170707) - NTRK1 (NM_002529) rearrangement: c.514-1686:LMNA_c.1251+18:NTRK1del Protein fusion: in frame (LMNA-NTRK1)	NA
EVT-P-0007739-T01-I M5-19	P-0007739-T01-IM5		LMNA-NTRK1	LMNA	unknown	threep_rime	156845035	12	452	True	retained	1.0	False	Protein kinase domain	PF13855; P F13895		LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exons 1-8 fused to NTRK1 exons 12-17): c.1443:LMNA_c.1354+235:NTRK1 Protein fusion: mid-exon (LMNA-NTRK1)	NA
EVT-P-0003705-T01-I M5-24	P-0003705-T01-IM5		LMNA-NTRK1	LMNA	unknown	threep_rime	156845433	12	492	False	retained	1.0	False	Protein kinase domain	PF13855; P F13895		c.686:LMNA_c.1476:NTRK1del Protein fusion: mid-exon (LMNA-NTRK1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0003329-T04-I M6-25	P-0003329-T04-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-12:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0003329-T03-I M6-26	P-0003329-T03-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-12:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0066239-T01-I M7-27	P-0066239-T01-IM7		LMNA-NTRK1	LMNA	unknown	three_prime	156845170	12	452	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0003329-T02-I M6-28	P-0003329-T02-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0023994-T06-I M6-29	P-0023994-T06-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	11	418	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0023994-T05-I M6-30	P-0023994-T05-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	11	418	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0023994-T01-I M6-31	P-0023994-T01-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	11	418	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-10 fused to NTRK1 exons 11-17): c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 5031-T02-I M6-32	P-0025 031-T0 2-IM6		LTAP1 -NTRK1	LTA P1	unknown	three_prime	156846 272	14	571	False	disrupted	0.7814814 81481481 4	True		PF1 385 5; P F13 895	Protein kinase domain	C1orf43 (NM_001098616) - NTRK1 (NM_002529): c. 168+114:C1orf43_c.1713: NTRK1inv Protein Fusion: mid-exon {C1orf43:NTRK1}	NA
EVT-P-002 4231-T01-I M6-33	P-0024 231-T0 1-IM6		MEF2 D-NTRK1	MEF 2D	in-frame	five_prime	156845 756	12	501	True	lost	0.0	False	PF138 55; PF 13895	Protein kinase domain		NTRK1 (NM_002529) - MEF2D (NM_005920) rearrangement: c.1502-11 6:NTRK1_c.1006+1908:M EF2Binv Protein Fusion: in frame {NTRK1:MEF2D}	NA
EVT-P-004 5731-T01-I M6-35	P-0045 731-T0 1-IM6		NOS1 AP-NTRK1	NOS1A P	unknown	three_prime	156843 455	8	294	False	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		NOS1AP (NM_014697) - NTRK1 (NM_002529) fusion: c.1106-363:NOS1A P_c.881:NTRK1dup Protein Fusion: mid-exon {NOS1AP:NTRK1}	NA
EVT-P-002 6354-T01-I M6-36	P-0026 354-T0 1-IM6		NTRK1 -ARHGEF11	ARH GEF11	in-frame	three_prime	156845 184	12	452	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		ARHGEF11 (NM_198236) - NTRK1 (NM_002529) rearrangement: c.4631-28 5:ARHGEF11_c.1355-128: NTRK1inv Protein Fusion: in frame {ARHGEF11:NTRK1}	NA
EVT-P-003 5466-T01-I M6-37	P-0035 466-T0 1-IM6		NTRK1 -ATP1A2	ATP 1A2	unknown	three_prime	156844 714	11	423	False	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		ATP1A2 (NM_000702) - NTRK1 (NM_002529) rearrangement: c.2563+2: ATP1A2_c.1268:NTRK1dup Protein Fusion: mid-exon {ATP1A2:NTRK1}	NA
EVT-P-005 8090-T04-I M7-38	P-0058 090-T0 4-IM7		NTRK1 -CADM3	CAD M3	out-of-frame	five_prime	156836 298	3	120	True	lost	0.0	False		Protein kinase domain	PF138 55	NTRK1 (NM_002529) - CADM3 (NM_021189) rearrangement: c.360-404: NTRK1_c.88+5312:CADM 3del Protein Fusion: out of frame {NTRK1:CADM3}	NA
EVT-P-002 1262-T01-I M6-41	P-0021 262-T0 1-IM6		NTRK1 -F11R	F11 R	in-frame	three_prime	156844 288	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		F11R (NM_016946) - NTRK1 (NM_002529) rearrangement: c.389-101: F11R_c.1196-75:NTRK1inv Protein Fusion: in frame {F11R:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 2467-T01-I M6-42	P-0022 467-T0 1-IM6		NTRK1 -IRF2B P2	IRF 2BP 2	in-frame	three_prime	156844 129	9	393	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) fusion (IRF2BP2 exon 1 fused with NTRK1 exons 9-17): c.1001-193:IRF2BP 2_c.1178-46:NTRK1inv Protein Fusion: in frame {IRF2BP2:NTRK1}	NA
EVT-P-001 0628-T01-I M5-43	P-0010 628-T0 1-IM5		NTRK1 -IRF2B P2	IRF 2BP 2	unknown	three_prime	156844 224	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement : c.948:IRF 2BP2_1195+32:NTRK1inv Protein fusion: mid-exon (IRF2BP2-NTRK1)	NA
EVT-P-001 0628-T03-I M6-44	P-0010 628-T0 3-IM6		NTRK1 -IRF2B P2	IRF 2BP 2	unknown	three_prime	156844 223	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement: c.950:IRF2 BP2_c.1195+31:NTRK1inv Protein Fusion: mid-exon {IRF2BP2:NTRK1}	NA
EVT-P-002 7278-T01-I M6-45	P-0027 278-T0 1-IM6		NTRK1 -KIF21 B	KIF2 1B	unknown	three_prime	156843 723	8	383	False	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		KIF21B (NM_001252100) - NTRK1 (NM_002529) rearrangement: c.2077+60 1:KIF21B_c.1149:NTRK1in v Protein Fusion: mid-exon {KIF21B:NTRK1}	NA
EVT-P-005 6197-T01-I M6-46	P-0056 197-T0 1-IM6		NTRK1 -NELF CD	NEL FCD	out-of-frame	five_prime	156843 148	7	284	True	lost	0.0	False	PF138 55; PF 13895	Protein kinase domain		NTRK1 (NM_002529) - NELFCD (NM_198976) rearrangement: t(1;20)(q23 .1;q13.32)(chr1:g.1568431 48::chr20:g.57565762) Protein Fusion: out of frame {NTRK1:NELFCD}	NA
EVT-P-002 7277-T01-I M6-79	P-0027 277-T0 1-IM6		NTRK1 -PEAR 1	PEAR 1	in-frame	three_prime	156844 219	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466:PEAR1 _c.1195+27:NTRK1dup Protein Fusion: in frame {PEAR1:NTRK1}	NA
EVT-P-002 9892-T05-I M6-80	P-0029 892-T0 5-IM6		NTRK1 -PLEK HA6	PLE KHA 6	in-frame	three_prime	156844 239	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKH A6_c.1195+47:NTRK1inv Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-002 9892-T04-I M6-81	P-0029 892-T0 4-IM6		NTRK1 -PLEK HA6	PLE KHA 6	in-frame	three_prime	156844 239	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKH A6_c.1195+47:NTRK1inv Protein Fusion: in frame {PLEKHA6:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 9892-T03-I M6-82	P-0029 892-T0 3-IM6		NTRK1 -PLEK HA6	PLE KHA 6	in-frame	three_prime	156844 239	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKH A6_c.1195+47:NTRK1inv Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-003 8819-T01-I M6-83	P-0038 819-T0 1-IM6		NTRK1 -RAB2 5	RAB 25	out-of-frame	five_prime	156844 458	10	417	True	lost	0.0	False	PF138 55; PF 13895	Protein kinase domain		NTRK1 (NM_002529) - RAB25 (NM_020387) Rearrangement : Protein Fusion: out of frame {NTRK1:RAB25}	NA
EVT-P-001 5004-T01-I M6-84	P-0015 004-T0 1-IM6		NTRK1 -RABG AP1L	RAB GAP 1L	out-of-frame	five_prime	156844 972	11	452	True	lost	0.0	False	PF138 55; PF 13895	Protein kinase domain		NTRK1 (NM_002529) - RABGAP1L (NM_014857) rearrangement: c.1354+17 2:NTRK1_c.2340+50594:R ABGAP1Ldel Protein fusion: out of frame (NTRK1-RABGAP1L)	NA
EVT-P-003 8568-T01-I M6-85	P-0038 568-T0 1-IM6		NTRK1 -SEMA 4A	SE MA4 A	out-of-frame	five_prime	156841 963	7	284	True	lost	0.0	False	PF138 55; PF 13895	Protein kinase domain		NTRK1 (NM_002529) - SEMA4A (NM_001193300) rearrangement: c.850+416: NTRK1_c.301-270:SEMA4 Adup Protein Fusion: out of frame {NTRK1:SEMA4A}	NA
EVT-P-004 8896-T01-I M6-86	P-0048 896-T0 1-IM6		NTRK1 -SMYD 2	SM YD2	unknown	five_prime	156844 382	10	405	False	lost	0.0	False	PF138 55; PF 13895	Protein kinase domain		NTRK1 (NM_002529) - SMYD2 (NM_020197) rearrangement: c.1215:NT RK1_c.602+876:SMYD2de l Protein Fusion: mid-exon {NTRK1:SMYD2}	NA
EVT-P-002 3987-T01-I M6-88	P-0023 987-T0 1-IM6		NTRK1 -TPR	TPR	unknown	three_prime	156844 285	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPR (NM_003292) - NTRK1 (NM_002529) rearrangement: c.2902:TP R_c.1196-78:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-002 4549-T03-I M6-89	P-0024 549-T0 3-IM6		NTRK1 -TPR	TPR	in-frame	three_prime	156844 290	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+651_c.1196-73inv Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-000 7050-T01-I M5-90	P-0007 050-T0 1-IM5		NTRK1 -VANG L2	VAN GL2	unknown	three_prime	156845 141	12	452	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		VANGL2 (NM_020335) - NTRK1(NM_002529) rearrangement : c.-191+44 56:VANGL2_c.1355-171:N TRK1dup Transcript fusion (VANGL2-NTRK1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 6172-T01-I M5-91	P-0006 172-T0 1-IM5		P2RY8 -NTRK1	P2RY8	unknown	five_prime	156838012	5	182	False	lost	0.0	False	PF13855	Protein kinase domain; P F13895		P2RY8(NM_178129) - NTRK1 (NM_002529) rearrangement: t(X;1)(chrX:g.1598513::chr1:g.156838012) Protein fusion: mid-exon (NTRK1-P2RY8)	NA
EVT-P-004 2526-T02-I M6-92	P-0042 526-T0 2-IM6		PRCC -NTRK1	PRCC	unknown	three_prime	156834591	3	120	False	retained	1.0	False	Protein kinase domain; PF13895		PF13855	PRCC (NM_005973) - NTRK1 (NM_002529) fusion: c.1323+10:PRCC_c.359:NTRK1del Protein Fusion: mid-exon {PRCC:NTRK1}	NA
EVT-P-004 4467-T01-I M6-93	P-0044 467-T0 1-IM6		SCP2 -NTRK1	SCP2	unknown	three_prime	156844773	11	443	False	retained	1.0	False	Protein kinase domain	PF13855; P F13895		SCP2 (NM_002979) - NTRK1 (NM_002529) fusion: c.1338+1979:SCP2_c.1327:NTRK1del Protein Fusion: mid-exon {SCP2:NTRK1}	NA
EVT-P-003 6662-T01-I M6-94	P-0036 662-T0 1-IM6		SHPRH -NTRK1	SHPRH	out-of-frame	five_prime	156831180	1	71	True	lost	0.0	False		Protein kinase domain; P F13855; PF13895		NTRK1 (NM_002529) - SHPRH (NM_001042683) rearrangement: t(1;6)(q231;q24.3)(chr1:g.156831180::chr6:g.146225530) Protein Fusion: out of frame {NTRK1:SHPRH}	NA
EVT-P-005 5621-T05-I M6-95	P-0055 621-T0 5-IM6		SLAMF6 -NTRK1	SLAMF6	out-of-frame	three_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain; PF13895		PF13855	SLAMF6 (NM_052931) - NTRK1 (NM_002529) fusion: c.50-1837:SLAMF6_c.360-291:NTRK1inv Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-005 5621-T02-I M6-96	P-0055 621-T0 2-IM6		SLAMF6 -NTRK1	SLAMF6	out-of-frame	three_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain; PF13895		PF13855	SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291:NTRK1inv Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-005 5621-T01-I M6-97	P-0055 621-T0 1-IM6		SLAMF6 -NTRK1	SLAMF6	out-of-frame	three_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain; PF13895		PF13855	SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291:NTRK1inv Protein Fusion: out of frame {SLAMF6:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 2936-T01-I M6-98	P-0022 936-T0 1-IM6		STK11 -NTRK 1	STK11	unknown	five_prime	156848955	15	616	False	disrupted	0.38888888 888888889	True	PF13855; PF13895		Protein kinase domain	NTRK1 (NM_002529) - STK11 (NM_000455) rearrangement: t(1;19)(;)(c hr1:g.56848955::chr19:g.204186) Protein Fusion: mid-exon {NTRK1:STK11}	NA
EVT-P-002 4875-T01-I M6-99	P-0024 875-T0 1-IM6		TAF A2 -NTRK 1	TAF A2	unknown	three_prime	156836548	4	120	True	retained	1.0	False	Protein kinase domain; PF13895		PF13855	NTRK1 (NM_002529) rearrangement: t(1;12)(q23.1;q14.1)(chr1:g.156836548::chr12:g.62314748) Transcript Fusion {FAM19A2:NTRK1}	NA
EVT-P-006 9757-T01-I M7-100	P-0069 757-T0 1-IM7		TARS2 -NTRK 1	TARS2	unknown	three_prime	156838028	5	187	False	retained	1.0	False	Protein kinase domain; PF13895	PF13855		TARS2 (NM_025150) - NTRK1 (NM_002529) fusion: c.264-188:TARS2_c.561:NTRK1del Protein Fusion: mid-exon {TARS2:NTRK1}	NA
EVT-P-005 0767-T01-I M6-101	P-0050 767-T0 1-IM6		TPM3 -NTRK1	TPM3	in-frame	three_prime	156845252	12	452	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		TPM3(NM_153469) - NTRK1 (NM_002529) Fusion(TPM3 exon 12 with NTRK1 exon12): c.665-654:TPM3_c.1355-60:NTRK1 inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-004 9078-T04-I M6-102	P-0049 078-T0 4-IM6		TPM3 -NTRK1	TPM3	unknown	three_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain	PF13855; PF13895		TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-004 9078-T01-I M6-103	P-0049 078-T0 1-IM6		TPM3 -NTRK1	TPM3	unknown	three_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain	PF13855; PF13895		TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-004 6877-T02-I M6-104	P-0046 877-T0 2-IM6		TPM3 -NTRK1	TPM3	in-frame	three_prime	156843811	9	393	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.776-357:TPM3_c.1177+60:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 6375-T04-I M6-105	P-0026 375-T0 4-IM6		TPM3 -NTRK1	TPM3	in-frame	three_prime	156844046	9	393	True	retained	1.0	False	Protein kinase domain	PF13855; PF13895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.775+355:TPM3_c.1178-129:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA

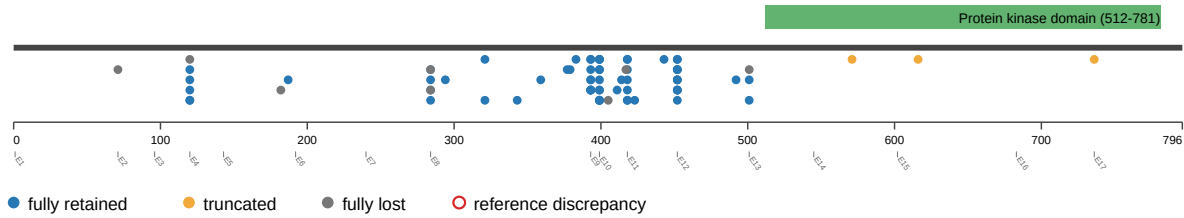
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 6375-T01-I M6-106	P-0026 375-T0 1-IM6		TPM3- NTRK1	TPM 3	in-frame	three_prime	156844 046	9	393	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 9-17): c.775+355:TPM3_c. 1178-129:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 5815-T02-I M6-107	P-0025 815-T0 2-IM6		TPM3- NTRK1	TPM 3	in-frame	three_prime	156844 230	10	399	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38: NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 1467-T01-I M6-108	P-0021 467-T0 1-IM6		TPM3- NTRK1	TPM 3	in-frame	three_prime	156844 038	9	393	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3 _c.1178-137:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 1170-T03-I M6-109	P-0021 170-T0 3-IM6		TPM3- NTRK1	TPM 3	in-frame	three_prime	156845 218	12	452	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_ c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-002 1170-T02-I M6-110	P-0021 170-T0 2-IM6		TPM3- NTRK1	TPM 3	in-frame	three_prime	156845 218	12	452	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_ c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-001 4143-T01-I M5-111	P-0014 143-T0 1-IM5		TPM3- NTRK1	TPM 3	in-frame	three_prime	156844 002	9	393	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM 3_c.1178-173:NTRK1inv Protein fusion: in frame (TPM3-NTRK1)	NA
EVT-P-006 9266-T01-I M7-112	P-0069 266-T0 1-IM7		TPM3- NTRK1	TPM 3	unknown	three_prime	156843 601	8	343	False	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c. .1027:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 9951-T01-I M6-113	P-0049 951-T0 1-IM6		TPR-N TRK1	TPR	in-frame	threep _prime	156844 660	11	418	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c. 1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-004 2965-T01-I M6-114	P-0042 965-T0 1-IM6		TPR-N TRK1	TPR	unknown	threep _prime	156843 703	8	377	False	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c. 1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-006 5034-T02-I M7-115	P-0065 034-T0 2-IM7		TPR-N TRK1	TPR	unknown	threep _prime	156843 651	8	359	False	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c. 1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-001 6347-T01-I M6-116	P-0016 347-T0 1-IM6		TRIM6 3-NTR K1	TRIM63	in-frame	threep _prime	156844 048	9	393	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TRIM63 (NM_032588) - NTRK1 (NM_002529) Rearrangement : c.1051+3 86:TRIM63_c.1178-127:N TRK1inv Protein Fusion: in frame {TRIM63:NTRK1}	NA
EVT-P-000 7755-T02-I M5-117	P-0007 755-T0 2-IM5		TRPM 8-NTR K1	TRPM8	out-of-frame	threep _prime	156845 664	13	501	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		TRPM8 (NM_024080) - NTRK1 (NM_002529) rearrangement: t(1;2)(q23.1; q27.1)(chr1:g. 156845664::chr2:g.234866 124) Protein fusion: out of frame (TRPM8-NTRK1)	NA
EVT-P-000 0104-T02-I M6-118	P-0000 104-T0 2-IM6		ZBTB7 B-NTR K1	ZBTB7B	unknown	threep _prime	156845 173	12	452	True	retained	1.0	False	Protein kinase domain	PF1 385 5; P F13 895		ZBTB7B (NM_001252406) - NTRK1 (NM_002529) Rearrangement : c.944:ZB TB7B_c.1355-139:NTRK1 del Protein fusion: mid-exon (ZBTB7B-NTRK1)	NA

Figures

domain retention outliers

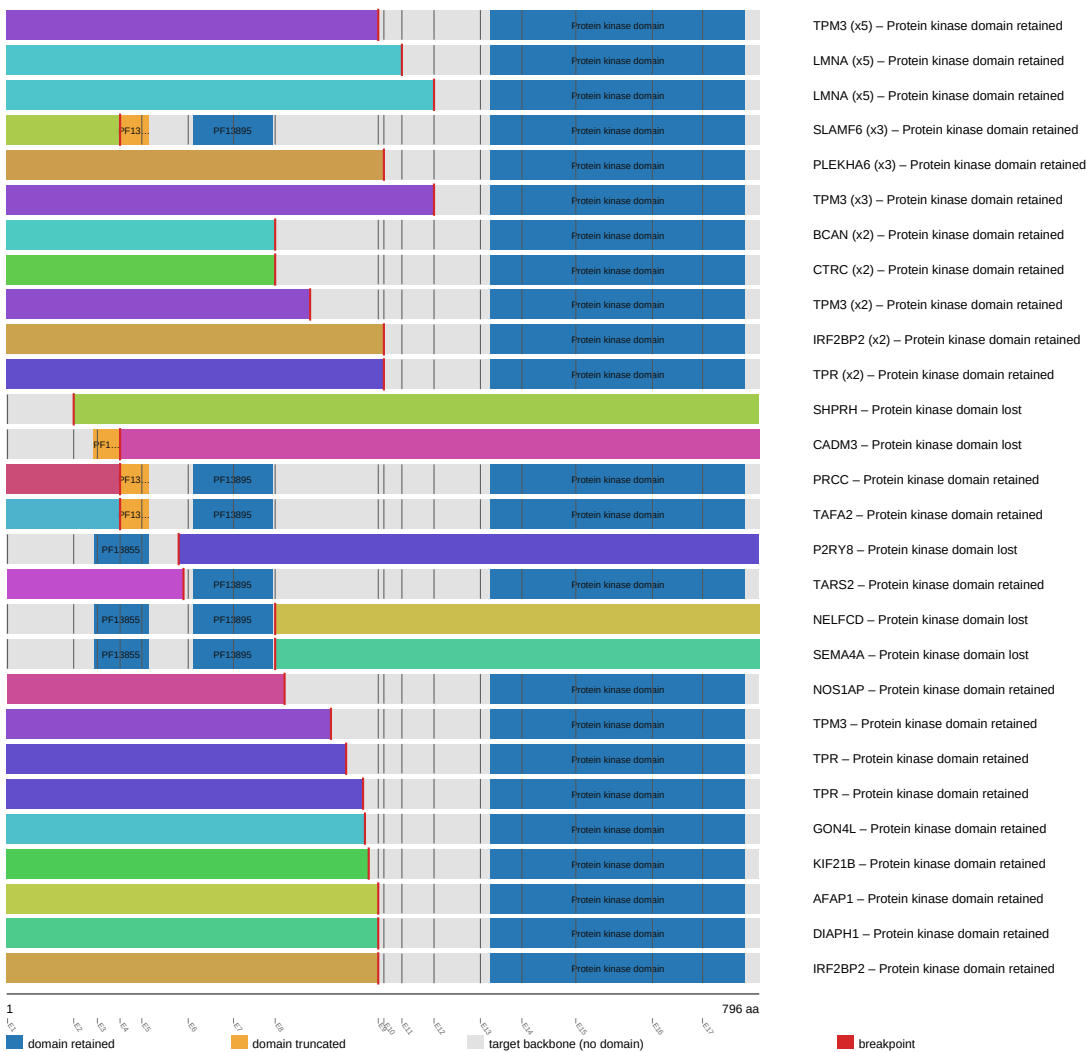
NTRK1 domain-retention track



fusion schematic

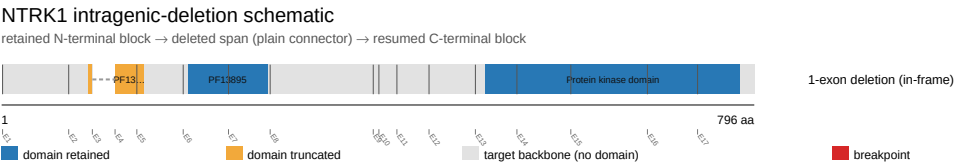
NTRK1 fusion-transcript schematic

partner block → breakpoint → retained NTRK1 portion (domain-colored, exon ticks)



Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

